

TECHNICAL SPECIFICATION FOR OUTDOOR TYPE DISTRIBUTION FEEDER PILLAR BOX

1. Scope:

This specification covers technical requirements of design, manufacture, testing at manufacturer's works, packing, forwarding, supply and unloading at store/site and performance of pillar box with all accessories for trouble free and efficient operation. This outdoor pillar box will be utilised for providing/interconnecting MVAC distribution network.

2. Applicable Standards:

The equipment covered with latest editions of the following standards/IEC:

- a) **IS 5039-1983:** Specification for distribution pillars for voltages not exceeding 1000V AC & 1200V DC.
- b) **IS 13703 (Part-1&2)-1993 / IEC 60263/1-1986:** Specification for LV fuses for voltage not exceeding 1000V AC & 1500V DC.
- c) **IS 12063-1987 / IEC 60529:** Classification of degree of protection provided by enclosure of electrical equipment.
- d) **IS 13947 (Part-1) / IEC 60947/1-1988:** Specification for LV switchgear & controlgear.
- e) **IS 8623 (Part-2)-1993 / IEC 60439/2-1987:** Specification for LV switchgear & controlgear assemblies – particular requirements for Busbar trunking system.
- f) **IS 2551-1982:** Danger Notice Plates.
- g) **IEC 60664:** Insulation co-ordination within low voltage systems including clearance creepage distances for equipment.
- h) **IS 2147:** Enclosure & Degree of protection of the Pillar Box
- i) **IS 6005:** Code of Practice for phosphating iron and steel
- j) **IEC 60269 :** HRC Fuses

3. Climatic Conditions of the Installation:

- a) Max. Ambient Temperature : 42 deg. C
- b) Min. Ambient Temperature : 10 deg. C
- c) Max. Humidity : 98%
- d) Min. Humidity : 60%

- e) Average Annual Rainfall : 2500 mm
 f) Average No. of rainy days per annum : 150 days
 g) Altitude above MSL : 9 metres
 h) Average wind velocity : 18 km/hr.

4. General Technical Requirement:

Sl. No.	Description	Requirement
1.	Installation	Outdoor
2.	Rated Voltage	430V
3.	No. of Phases	3
4.	Frequency	50 Hz
5.	Wiring System	3 Phase 4 Wire
6.	System Earthing	Solid at source end.
7.	Impulse withstand voltage	8 kV
8.	Rated short time fault current	50 kA for 1 sec.
9.	Busbar Material	
	6 way (2 I/C + 4 O/G)	Aluminium (75 X 12) mm dimension.
	8 way (2 I/C + 6 O/G)	Aluminium (100 X 12) mm dimension.
10.	Max. current density of AlBusbar	1 Amp/sqmm.
11.	Degree of Protection	IP-43
12.	Temperature Rise	Max. permissible temperature rise for the Busbar and terminal should be 45°C & 65°C respectively at ambient not exceeding 42°C
13.	No. of Incoming unit with solid link	2
14.	No. of Outgoing unit with fuse link	4 for 6 Way, 6 for 8 Way Pillar box
15.	Type of Incoming Fuse-Link / Fuse Base	Knife type Solid Links with tinned copper contacts & fuse base with high stability glass fibre reinforced thermosetting polyester.
16.	Type of Outgoing Fuse-Link / Fuse Base	Knife type DIN HRC Fuses with tinned copper contacts & fuse base with high stability glass fibre reinforced thermosetting polyester.
17.	Type of Incoming Solid-Link / Base	Knife type solid link with tinned copper contacts & base with high stability glass fibre reinforced thermosetting polyester. However for 4 Way PB there won't be any solid link and all the units should have fuse link.
18.	Fuse Base Rating & Size	
	6 way	630 Amps (Size-3)
	8 way	1000 Amps
19.	DIN HRC fuse Rating & Size	315 Amps (Size-3)/400 Amps (Size-3) depending upon site selection.

20.	Current rating of Busbar	
	6 way (2 I/C + 4 O/G)	900 Amps
	8 way (2 I/C + 6 O/G)	1200 Amps
21.	Busbar spacing (centre to centre)	185 mm
22.	Neutral Busbar rating	
	6 way (2 I/C + 4 O/G)	900 Amps
	8 way (2 I/C + 6 O/G)	1200 Amps
23.	Size of Incoming Cables	2X 4C X 300 sqmm.
24.	Size of Outgoing Cables	4C X 185 sqmm.
25.	Pillar box dimension	1400 X 1200 X 415 mm.
	6 way (2 I/C + 4 O/G)	1400 X 1400 X 415 mm.
	8 way (2 I/C + 6 O/G)	1400 X 1200 X 415 mm.
26.	PB Leg dimension	910 mm X 4 (made of 75X75X 6 mm angle iron)
27.	Enclosure material	Cold rolled steel of 3.15 mm thickness
28.	Colour of enclosure	Light Grey 631 of IS:5

NB.: Suitable **Correction factor and Diversity factor as per requirement of IEC/EN 61439-2** are to be followed during designing the assembly with in the PB enclosure.

5. General Construction:

- The design and layout of pillar box shall be compact and convenient to provide ease of termination of cables and operation of fuse bases. It shall be designed for termination of cables and operation of fusebase shall be from the front end. The Feeder Pillar enclosure shall provide a degree of protection not less than IP-43 in accordance with IS-2147.
- The enclosure shall be suitable for Outdoor installation. The enclosure shall be made from cold rolled sheet steel of thickness 3.15 mm and shall be suitable to withstand rough usage and weather.
- Panels shall be rigid free standing & floor mounting type and comprise of structural frames enclosed completely with specially selected smooth finished cold rolled sheet steel of thickness not less than 3.15 mm for weight bearing members of the panels such as front sheet and door frames, sides, door top and bottom portions. The Angle structure for Leg 75x75x6mm.
- All joints to be made flush type and all edges shall be bent at right angles and rounded. All structural members shall be bolted or welded together. Necessary arrangement shall be provided for bolting together the adjacent panels. The doors, removable covers and panels shall be sealed with gasket all round with neoprene gaskets. Ventilating louvers if provided shall have screens and filters. **The panels shall be designed for reception of external control**

cables (if any) from the bottom and shall have detachable gland plates split in two halves. Cable gland plate, fitted on the bottom of panel shall be connected to earthing of panel.

- e) The enclosure shall be sized to facilitate easy maintenance at minimum height of 300 mm above FFL.
- f) The top of pillar shall be fitted with a sloping canopy, the design of which shall be such that rain water shall not accumulate on the top.
- g) It shall have overhang and fascia of min. 100mm. 2 nos. lifting lugs of same material shall be provided.
- h) Three nos. heavy duty hinges shall be provided for each door such that they are not visible from outside and hence not removable. The hinge shall be of such construction that the door can be swung open by not less than 150°. The door shall be fitted with integral handle, flushed with the surface of the door.
- i) Air ventilation arrangement shall be provided at top and all sides of the enclosure to facilitate air circulation.
- j) The door shall have three point locking arrangement. Special type tamperproof heavy duty built-in inter-lock with appropriate key shall be provided. Locking system with handle up & down arrangement shall be provided on front side of the doors. The door shall be provided with pad-locking facility also.
- k) The Pillar shall be provided with removable split type base plate fitted with rubber grommets for cable entry to facilitate cable jointing.
- l) Proper support shall be provided for holding I/C& O/G cables. These, cable support arrangement shall be provided to take the load of the cable.
- m) The Busbar of all phases and neutral shall be provided with colour coded heat shrinkable insulation sleeves. Phase barriers shall be provided between the Busbars and all Busbars shall be in two equal sections with removable link.
- n) Cable termination arrangement shall be from bottom and suitable for external cables of type and sizes as mentioned in the specification. Additional arrangement shall be provided with busbar for terminating cable for DG Set in emergency. The pillar shall also be provided with removable split type side plate to facilitate cable entry and termination for this purpose.
- o) The pillar box shall be provided with a LED lamp rated for 230V, 1 Phase, 50 Hz supply for illumination inside the enclosure. A 230V, 1phase, 50 Hz, 5/15 Amps, 5 pin receptacle with switch shall be provided inside the enclosure at a convenient location.
- p) The design of pillar box shall ensure there is no possibility of the operator experience a shock during normal operation. Insulated barriers shall be provided wherever necessary so as to ensure that no accidental contact with live parts inside the pillar box is possible.
- q) The Pillar box shall be provided with an Aluminium Earth bus at the bottom, suitable for the rated short time current of the pillar. The earth bus shall provide Earthing to all the components inside the pillar and the metallic enclosure including the armour of the cables.

The earth bus shall be internally connected to two separate earth terminals located outside, on either side of the pillar. The Busbar shall not be visible or removable from the outside of the cubicle. Door Earthing shall also be provided with bolted lugs at both ends. The neutral shall be insulated from the earth.

- r) The paint on the pillar box shall have good colour retention properties and shall be weather and UV resistant. The Retro reflective paint strip shall be provided on the four faces of the pillar box which shall withstand the temperature conditions and reflect vehicular light at night time providing safety to traffic. All sheet steelwork shall be phosphated in accordance with IS: 6005 "Code of practice for phosphating iron and steel". Oil, grease, dirt and swart shall be thoroughly removed by emulsion cleaning. Rust and scale shall be removed by pickling with dilute acid followed by washing with running water rinsing with slightly alkaline hot water and drying. After phosphating, thorough rinsing to be carried out with clean water followed by final rinsing with diluter dichromate solution. After that the powder spray with spray gun to be used and to put into the oven for drying in 200deg.C. for 40min
- s) The pillar box shall be provided with durable and legible Aluminium nameplates containing all technical parameters and manufacturing details. A plate of statutory instruction like 'stick no bill' shall be provided on all four side of the pillar box. The pillar box shall be provided with a plate 'Danger' mark for 430V AC as per IS 2551-1982. Property plate mentioning "Property of WBSEDCL" and project name if any shall also be provided.
- t) All routine, acceptable & type tests shall be carried out in accordance with the relevant IS/IEC. All routine/acceptance tests shall be witnessed by the Purchaser/his authorized representative.
- u) Equipment shall be subject to pre-dispatch inspection by a duly authorized representative of the Purchaser. Inspection may be made at any stage of manufacture at discretion of the purchaser and the equipment, if found unsatisfactory as to workmanship or material, the same is liable to rejection.
- v) The material received at Purchaser store will be inspected for acceptance and shall be liable for rejection, if found different from the reports of the pre-dispatch inspection.
- w) Supplier shall stand guarantee for five years from the date of supply towards design, materials, workmanship and quality of process/manufacturing of items under this contract.
- x) Supplier shall ensure that all the equipment covered under this specification shall be prepared for transport in a manner so as to protect the equipment from damage in transit.
- y) Supplier shall submit with the offer Quality assurance plan indicating the various stage of inspection, the tests and checks which will be carried out on the material of construction, components during manufacturing and equipment after finishing.
- z) Supplier shall provide Instruction Manuals covering erection & maintenance instruction and all relevant information & drawings with complete BOM pertaining to main equipment and also list of complete set of accessories and tools required for erection and maintenance with each of the pillar boxes.

6. Special Requirement:

- a) Fuse blown indication with LED for Easy identification for the operations which phase fuse has blown, saving time and operation of different poles.
- b) Mechanism of Fuse removal without touching fuse so that there is no need to touch blown fuse which is quite hot.
- c) Stainless steel spring material at contacts so that material corrosion is avoided and also losses/temperature at contacts are reduced.
- d) Wider Spacing at terminals for connecting cable so that it is easy for operators to connect /disconnect cables.

7. TESTS:

7.1 TYPE TESTS: ON COMPLETE PILLER BOX:

The equipment offered in the tender should have been successfully type tested at NABL laboratories in India or equivalent international laboratories in line with the relevant standard and technical specification, within the last 5 (five) years from the date of offer. The bidder shall be required to submit complete set of the type test reports along with the offer.

The list of type tests is as follows:

Following type tests as per IS: 5039/1983 shall be carried out on complete L.T. feeder Pillars.

- 1. Temperature rise test:-The temperature rise test should be carried out as per IS: 8623 –1993.
- 2. High voltage test shall be carried out as per IS: 8623/ 1993 amended up to date.
- 3. Short Time Withstand Current Test on L.T. Pillar shall be carried out as per IS 8623: or latest version.
- 4. Degree of protection for IP- 43 on complete L.T. Pillar shall be carried out as per IS: 13947/1993 or the latest version thereof.
- 5. Verification of clearance & creepage distances.

7.2 ACCEPTANCE TEST: ON COMPLETE PILLER BOX:

Following tests shall be carried out as per acceptance tests in addition to routine tests on one random sample of each rating out of the lot offered for inspection:

- i) Temperature rise test on one sample of each rating.

7.3 ROUTINE TESTS: ON COMPLETE PILLER BOX

- a) Routine and acceptance tests as per relevant IS shall be carried out at the manufacturers' works.
- b) Overall Dimensions Checking.
- c) Insulation Resistance Tests.
- d) High Voltage Test at 2500 V, 50 Hz AC for one minute.

8. INSPECTION:

- 8.1. The supplier shall prepare a prototype and offer the same for inspection and approval before taking up mass production.
- 8.2. All routine & acceptance tests and inspection of material shall be carried out at the place of manufacturer. The manufacturer shall offer the Inspector (representing the purchaser) all reasonable facilities, free of charge at the time of inspection.
- 8.3. The supplier shall offer inspection of the material at his works before dispatch. If required the supplier shall also offer inspection of manufacturing painting and assembly processes and quality control system. If any material is not found in line with relevant specification the supplier shall carry out the modification and offer for re-inspection.
- 8.4. Inspection and acceptance shall not absolve the supplier of his responsibility to supply the material in
- 8.5. accordance with the specifications. The purchaser reserves the right to reject the material not confirming the relevant specifications.

9. DRAWINGS AND DOCUMENTATION:

- a. The successful bidder shall submit set of all drawings of the L.T. Pillar Boxes and its components in
- b. triplicate and get the same approved before commencement of production.
- c. The tenderer shall furnish all details and clarifications required if any for scrutiny and evaluation of the offer.
- d. Manufacture of material to be supplied shall be done strictly as per approved drawing.
- e. Approval of drawing shall not absolve the supplier of his liability for ensuring correctness according to applicable standards & regulations.

10. DESPATCH:

- 10.1 The material to be supplied shall be packed and dispatched only after inspection and approval.
- 10.2 Supplier shall be responsible for packing, transporting and delivery to the consignee at anywhere of West Bengal.
- 10.3 Copies of packing list and inspection report duly approved shall be sent along with each consignment.

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